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18 April 1961
SUPERSEDING
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MILITARY SPECIFICATION

STERILIZERS, SURGICAL DRESSING; STERILIZERS, INSTRUMENT; AND STERILIZERS, WATER

(NAVAL SHIPBOARD)

1. SCOPE

1.1 Scope. - This specification covers surgical dressing, instrument and water sterilizers which are steam and electrically heated, self-contained, shockproof and have standard installation dimensions. The sterilizers are intended for Naval shipboard use.

1.2 Classification. - Sterilizers shall be of the following types, as specified (see 6.1):

Type I - Dressing sterilizers.
Type II - Instrument sterilizers, pressure type.
Type III - Instrument sterilizers, boiling type.
Type IV - Water sterilizers.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-R-175 - Resistance Wire.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-S-901 - Shockproof Equipment, Class HI (High-Impact), Shipboard Application, Tests for.
MIL-E-917 - Electric Power Equipment, Basic Requirements for (Naval Shipboard Use).
MIL-D-963 - Drawings, Electrical, Hull and Mechanical Equipment for, Naval Shipboard Use.
MIL-R-2033 - Relays for Naval Shipboard Electrical Service.
MIL-C-2174 - Controllers, Motor Starters, and Master Switches - Direct Current, Naval Shipboard Use.
MIL-C-2212 - Controllers, Motor Starters, and Master Switches - Alternating Current, Naval Shipboard Use.
MIL-M-15071 - Manual, Technical, for Mechanical and Electrical Equipment, (Less Electronics).
MIL-C-15987 - Cable Assemblies, Power, Electrical (With Grounding), for 440-Volt Portable Equipment, (Shipboard Use).
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurements funtions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN IRON AND STEEL INSTITUTE
Steel Products Manual

(Application for copies should be addressed to the American Iron and Steel Institute, 350 Fifth Avenue, New York 1, New York.)

AMERICAN STANDARDS ASSOCIATION

C33.1 - Flexible Cord and Fixture Wire, Safety Standard for.
C73a - Grounding- Type Attachment Plug Caps and Receptacles and 3-Wire, 30-Ampere, 250-Volt Cap and Receptacle.

(Application for copies should be addressed to the American Standards Association, 70 East 45th Street, New York 17, New York.)

FSC 6530

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OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 1, New York.)

3. REQUIREMENTS

3.1 Materials. - Materials shall be as specified hereinafter.

3.1.1 Corrosion-resisting steel shall conform to type 302 or 304 of the American Iron and Steel Institute, except parts that do not come in contact with water or steam such as finishing jackets, panels for recessed mounting and stands may be type 430 of the American Iron and Steel Institute.

3.1.2 Materials welded, connected or bolted together shall be those which do not cause electrolysis.

3.2 Piping, valves and fittings. -

3.2.1 All sterilizers that can be operated with steam as a heating source shall be complete with piping, valves, cocks, thermostatic traps, return check valves and fittings required for control of water supply and waste, steam supply and return. Piping valves and fittings shall be of copper, brass, bronze, nickel-copper alloy (monel) or corrosion-resisting steel. Exposed pipes, valves or pipe fittings of copper, brass or bronze shall be chromium or nickel plated. Plated parts shall have a polished finish.

3.2.2 The piping shall terminate at the control valves of water or steam supply lines, at the check valves on steam return lines, and at the open receiver on waste lines, except that for combination mountings, the pipes of each kind shall be connected and brought to a common pipe fitting practically flush with the stand. In all cases, a separate thermostatic trap shall be supplied for return from each steam heating coil, regardless of method of mounting.

3.2.3 Securing of piping and fittings. - All piping valves, steam coils, and fittings, shall be adequately secured in position so as to prevent them from vibrating, and also to reduce strain where connected to sheet metal.

3.2.4 Pipe sizes. - Piping for water supply, steam and return, shall be standard pipe size or tubing of a comparable capacity. Overflows, wastes, and vents may be tubing not less than 0.032 inch thick with slip joints.

3.2.5 Valves. - Valves shall have renewable discs with slipon holders and shall be tested to a hydrostatic pressure of 300 pounds per square inch gage (psig) on the shell and 150 psig on the seat. Valve handles shall be of heat-resistant plastic material. The function of the valve shall be indicated on the handle in a clear and durable manner.

3.2.6 Receivers. - Open receivers shall be of brass, bronze or copper.

3.3 Gages and thermometers. -

3.3.1 Water gages. - Water level indicating glasses shall be protected with guards finished similar to the sterilizer. These guards shall consist of metal rods parallel to the gage or a perforated metal cylinder surrounding the gage. The gage glass shall be shielded by a transparent, heat-resistant plastic tube.

3.3.2 Dial type gages. - Dial gages shall have a glass face over the dial. Cases shall be finished in polished chromium plate, polished nickel plate, or anodized aluminum alloy.

3.3.3 Thermometer range. - Thermometers shall be suitable for temperatures up to 280°F. They shall be easily read. Four °F. shall correspond to at least 1/16-inch linear scale distance.

3.3.4 Accuracy. - Pressure indicating gages shall be correct within 2 pounds per square inch (psi) over the entire range. Thermometers shall be correct within 2°F. over the entire range.

3.4 Steam-heating systems. -

3.4.1 Pressure. - When operating with steam as a heating source the sterilizers shall operate in accordance with the requirements of this specification on a minimum line pressure at 35 psig to a maximum line pressure of 75 psig at the sterilizer.

3.4.2 Coils and tubes. - Heating and cooling coils or tubes shall be seamless, annealed copper which is tinned on the outside with pure tin and tested to a hydrostatic pressure of 250 psig. Nipple and coil or nipple and tube steam connections shall be secured with brazed fittings, ground joint unions and locknuts, or be swaged and tested to a hydrostatic pressure of 250 psig.

3.4.3 Safety valves. - Safety valves shall be set at the proper pressure and locked so they may be unlocked only with tools. Safety valves shall be such that the operating pressure cannot rise above 25 psig and shall blow down at least 2 psig before shutting off. If two safety valves are furnished on one sterilizer, one shall operate 2 psi ahead of the other.

3.4.4 Strainer. - Steam strainers which effectually clean the steam of solid matter and which may

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be readily removed for cleaning shall be provided in steam-supply lines.

3.4.5 Steam trap. - Thermostatic steam traps shall be suitable for the operating pressure. They shall automatically pass air and condensation and partially close against steam, and shall be compensated for pressure and temperature. Each thermostatic return trap shall be supplied with check valve and shall have sufficient lead from the coil or chamber which it serves to cause it to cool sufficiently to pass a slight amount of vapor.

3.5 Electric heating systems. -

3.5.1 General requirements. - Current-carrying parts shall be enclosed and shall be mounted on plastic in accordance with Specification MIL-E-917. Sterilizers specified for operation on 440 volts, 3 phase shall have the load balanced equally on each phase.

3.5.2 Voltage. - The sterilizers shall operate on one of the following voltages, as specified (see 6.1):

(a) 115 volts, single phase, 60 cycle, alternating current (a.c.) (type III, size 2 only).
(b) 115 volts, direct current (d.c.).
(c) 230 volts, d.c.
(d) 440 volts, 3 phase, 60 cycle, a.c.

3.5.3 Dielectric. - Electrically heated sterilizers shall withstand for a period of one minute, a dielectric test voltage of twice rated voltage plus 1000 volts when tested as specified in 4.4.4.

3.5.4 Insulation resistance. - The insulation resistance of each electrically heated sterilizer when at normal room temperature shall be not less than 50 megohms. The insulation resistance of the sterilizer at normal operating temperature shall be not less than 10 megohms when tested as specified in 4.4.5.

3.5.5 Controls. -

3.5.5.1 Contactors and relays. - Magnetic contactors or relays shall be furnished as part of each type I, II or IV sterilizer. For 440 volts, 3 phase, 60 cycle, a.c. sterilizers, contactors conforming to Specification MIL-C-2212 shall be used. For d.c. sterilizers, contactors conforming to Specification MIL-C-2174 or relays conforming to Specification MIL-R-2033 shall be used. Momentary opeing or closing of the contacts under shock is permissible where operation of the sterilizer is not affected. On d.c. sterilizers, the contactor or relay shall break both sides of the line. On 3-phase sterilizers the contactor shall break each line.

3.5.5.2 Pressure switches. - Pressure switches shall conform to Specification MIL-E-917 for insulating materials and electrical creepage and clearance distances.

3.5.5.3 Switches. - Sterilizers shall be provided with reliable "ON-OFF" switches. The switch positions shall be clearly marked.

3.5.5.4 Low water cutouts and thermostatic switches. - Low water cutouts and thermostatic switches shall be reliable and rugged and shall operate by expansion of a metal diaphragm or by equally positive means. Cutouts shall open in case of excessive temperature and shall require manual reset.

3.5.6 Heating elements. - Heating elements shall be of the enclosed type. Heating elements shall consist of resistance wiring conforming to composition D or E of Specification QQ-R-175 embedded in refractory material, preferably magnesium oxide. The resistance wire and refractory material shall be encased in a metal sheath which shall firmly hold the wire and refractory material in place. The sheath of immersion heaters shall be tinned copper, nickel plated copper, nickel-copper alloy, corrosion-resisting steel or nickel-copper-iron alloy. The refractory material shall be tightly packed to prevent the wire from grounding to the sheath when subjected to bending, forming, shock, or vibration. At points where connections are made, a length shall be left unheated to prevent damage to the connections and wire by overheating. The ends of the heating elements shall be effectively sealed to prevent the entrance of moisture. Connections to the elements shall have locknuts or lockwashers. Heating elements shall be readily replaceable without disconnecting piping.

3.5.7 Indicator light. - Each electrically heated sterilizer shall be equipped with an indicator light having a red lens at least 1/2 inch in diameter. The indicator light bulb may be either a neon or incandescent type but shall be replaceable from the front. The light shall be mounted on the front of the sterilizer and shall be illuminated when the "ON-OFF" switch is in the "ON" position. When necessary, a resistor or transformer shall be used in the indicator light circuit.

3.5.8 Wiring. - All wiring shall be enclosed. Conduit shall be used for all wiring not otherwise enclosed.

3.5.8.1 Wire run in a heated zone exceeding 400°F., or connected to heating unit terminals where the wire terminal connection temperature exceeds 400°F., shall be nickel conductor not smaller than 4107 circular mils (AWG No. 14) with heavy asbestos insulation. If the wire is brazed or welded, the conductor shall be manganese-nickel (AWG No. 12).

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3.5.8.2 Wire carrying heater current, except that covered by 3.5.8.1, shall be tin coated copper stranded conductor not smaller than 3830 circular mils (approximately AWG No. 14) insulated with silicone rubber with an outer covering of glass braid.

3.5.8.3 Wiring for control circuits shall be tin coated copper stranded conductor not smaller than 1780 circular mils (approximately AWG No. 18) insulated with silicone rubber with an outer covering of glass braid.

3.6 Stands. - Sterilizers that are stand mounted shall be secured on integral stands of heavy seamless or butted corrosion-resisting steel tubing. Joints in stands shall be welded or brazed with joints of exposed stands ground smooth after welding or brazing. Stands shall rest on plated bronze floor plates. All welding and brazing shall be continuous.

3.7 Shock resistance. - Sterilizers shall withstand the HI shock tests of Specification MIL-S-901. The recording thermometers, is specified, (see 3.8.14.2) shall not be shock tested.

3.8 Type I, dressing sterilizers, direct steam and dual-voltage electric. -

3.8.1 Type I sterilizers shall have provision for being heated both from direct steam supply and from an electric supply. They shall have horizontal chambers suitably steam-jacketed. They shall operate at steam pressures of 15 to 20 psig in the sterilizer chamber, shall provide for complete saturation of contents of the chamber, and shall maintain a temperature throughout the chamber of not less than 250°F. Means shall be provided for exhausting chamber steam and drying a full load following sterilization. The drying may be vacuum process or non-vacuum process.

3.8.2 Sizes of type I sterilizers. - The sterilizers shall conform to table I for the size specified (see 6.1). The time required to reach sterilizing temperature (250°F) shall be based on water at 70°F and with water level within 1/2 inch of the top of the gage when the sterilizer is in an ambient room temperature of not more than 104°F. nor less than 78°F.

Table I - Dimensions and requirements for type I, dressing sterilizers.

Sizes	Dimensions of inside of chamber (Minimum)		Thickness of shells (Minimum)		Time required to attain sterilizing temperature (250°F., max.)		Power input (maximum)
	Diameter	Length	Inner chamber	Outer chamber	With direct steam heat	With electrical heat (at rated voltage)	
	Inches	Inches	Inch	Inch	Minutes	Minutes	Kw
2	16	24	0.093	0.078	10	20	11
3	20	36	.125	.078	15	30	14

3.8.3 Mounting of type I sterilizers. - Sterilizers shall be stand mounted with a corrosion-resisting steel panel adapted for open or recessed installation. Gages and operating handles shall be located in front of the panel.

3.8.4 Door opening. - The door opening shall be approximately the same diameter as the inside of the chamber and shall provide an unobstructed access to the chamber.

3.8.5 Body. - The body shall have two cylindrical horizontal chambers, one telescoping the other, and shall be suitably staged, forming a steam chamber not less than 9/16 inch deep over the horizontal wall of the inner chamber. The steam chamber may extend over the rear end as well as the horizontal walls of the chamber. If the rear end of the chamber is not steam jacketed, the opening at the end of the steam chamber shall be closed by a casting or dished plate to which the chamber walls shall be securely welded. A baffle plate of nickel-copper alloy (monel) or corrosion-resisting steel shall be fitted on the inside surface of the rear end so as to provide a steam space of not less than 1/4 inch between the head and baffle. The baffle plate shall be not more than 2-1/2 inches smaller in diameter than the diameter of the inner chamber. If the rear end is steam jacketed, no baffle plate shall be required.

3.8.5.1 Inner shells of sterilizers including backhead shall be of welded nickel-copper alloy (monel). The outer shell shall be of welded corrosion-resisting steel properly secured.

3.8.6 Door and collar. - The door and collar shall be of cast bronze, wrought nickel-copper alloy or other corrosion-resisting metal approved

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by the bureau or agency concerned. The collar shall be welded to the shells. Exposed bronze parts shall be nickel plated.

3.8.6.1 Doors shall have clamping levers operated by heat insulated handles or handwheel, shall fit against a synthetic rubber gasket, and shall be hinged to permit proper alignment of the door. Clamping levers, center stud and necessary parts in connection with same may be of steel. Parts of the locking device, except as previously specified, shall be of bronze. A safety mechanism shall be provided to assure the complete exhaust of internal pressure before the door may be opened.

3.8.6.2 A removable vertical baffle of corrosion-resisting steel or nickel-copper alloy (monel), shall leave a space for free circulation of steam not less than 1/4 inch in depth between the door and the contents of the sterilizer.

3.8.7 Drying of dressings. - Drying of dressings following sterilization may be accomplished by means of one of the following methods:

(a) Exhaust the steam from the sterilizer chamber to zero gage pressure, leaving the jacket pressure on to keep the walls of the chamber hot. When chamber pressure has been exhausted, unlock the door and open it slightly, about 1/4 inch, to discharge vapor.
(b) Exhaust the steam from the chamber to zero gage pressure, then by vacuum method to exhaust about 1/3 of the vapor. Then break the vacuum and open the door 1/4 inch to allow the remainder of the vacuum to exhaust with the vapor.
(c) Exhaust the steam from the chamber to zero gage pressure, then employ a vacuum dryer device which functions as a pump to circulate filtered air to the chamber where it entrains with the vapor and both are then discharged through a vent.

3.8.8 Steam generator. - Sterilizers shall be fitted with an independent steam generator constructed of corrosion-resisting steel.

3.8.8.1 The steam generator shall be of sufficient capacity for two complete sterilizations of 30 minutes each, plus 10 minutes drying, with one filling of water.

3.8.8.2 The steam generator shall have an end removable for cleaning. The end shall be held in place against a suitable gasket.

3.8.8.3 The complete steam generator shall withstand a hydrostatic pressure of 50 psi for 10 minutes, and shall show no leakage or sweating.

3.8.8.4 The steam generator shall be provided with a filling valve, draw-off valve, and water gage. The water gage shall not become airbound and shall positively identify the depth of the water in the steam generator.

3.8.9 Fittings. - The following fittings shall be provided:

(a) A thermostatic trap connected in the drain from the sterilizing chamber or connected to an opening on the lower side of the sterilizer front, so as to maintain a slight discharge and a constant circulation of steam during the entire sterilizing period. A steam strainer, or other means approved by the bureau or agency concerned, for preventing the trap from becoming clogged shall be provided.
(b) One steam pressure (and vacuum, if used) gage for the sterilizing chamber.
(c) One steam pressure gage for the outer jacket.
(d) A safety valve(s) set at 25 pounds pressure.
(e) A suitable spreader or deflector plate at point where steam enters the sterilizing chamber.
(f) Operating valve(s) mounted in an accessible place, with all functions plainly marked thereon.
(g) Indicating thermometer located near the sterilizer where readable. The calibrated graduation shall be from not more than 200 to at least 280°F. The thermometer bulb shall be located in the lowest part of the discharge steam line at its nearest junction with the sterilizer chamber.
(h) A check valve shall also be provided to close the leak port when drawing vacuum, if necessary.
(i) An open receiver furnished separately for connection by the installing activity.

3.8.10 Exhaust and drains. - The exhaust to atmosphere from the operating valve shall be provided with a drain. The drain, together with the drain from the sterilizing chamber, and the drain from the steam generator shall be piped to one or more open receivers.

3.8.11 Steam heating. - Sterilizers shall be capable of operation direct from steam lines. A thermostatic trap and check valve shall be supplied on return form the jacket. Each sterilizer shall be fitted with an automatic control which shall maintain the pressure within 2 psig of the set pressure, with a fluctuation in boiler pressure not to exceed 15 percent, and permit operation of the sterilizer at 15 to 20 psig within the sterilizing chamber.

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3.8.12 Electrical requirements.- Heating elements in the steam generator shall be of sufficient wattage to produce sterilizing temperature in the time specified but shall not exceed the maximum rating specified in 3.8.2 (see table I). A reliable low-water cutout shall be supplied to automatically cut off power to the heater when water in the steam generator is depleted.

3.8.12.1 Each sterilizer shall be fitted with an automatic control. The control shall function through a flexible bellows, actuated by the pressure of the sterilizer, or by means of a thermostatic switch, either of which shall have a range of adjustment from 15 to 20 psig. The desired pressure shall be automatically maintained with a total variation not to exceed 2 psig above or below the desired range.

3.8.13 Installation dimensions.- The installation dimensions shall be as shown on figures 1 and 2.

3.8.14 Recording thermometer.- When specified (see 6.1), a recording thermometer shall be supplied. Recording thermometers for sterilizers shall be fitted on the sterilizer or provided with fittings for wall mounting.

3.8.14.1 Recording thermometers shall be made so that the bulb may be readily installed in or detached from the equipment. Bulbs shall be made of a durable corrosion-resisting metal. The capillary tube shall be suitably protected and long enough to permit placing the bulb in a suitable fitting in the discharge line from the chamber, back of the indicating thermometer. The dial of the recording thermometer shall be not less than 8 inches in diameter. The recording element shall be enclosed in a dustproof case finished in black enamel, with glass front, hinged and secured with lock and key. The recording mechanism shall make one revolution in 24 hours. The record shall be easily readable. The recorder shall be complete with 100 charts, pen and ink, and other necessary accessories.

3.8.14.2 The HI shock provisions of 3.7 are not applicable to recording thermometers.

3.8.15 Accessories.- When specified (see 6.1), the accessories specified in 3.8.15.1 and 3.8.15.2 shall be furnished.

3.8.15.1 Containers for dressings.- Containers shall be approximately 8 inches deep. The difference between the inside diameter of the sterilizer and the outside diameter of the container shall not exceed 2-1/2 inches. Containers shall be constructed of corrosion-resisting steel and shall be not less than 0.028 inch thick, except that side walls shall be not less than 0.022 inch thick if strongly reinforced. They shall have ports, adequate to allow thorough penetration of steam, without sliding covers. Containers shall be arranged so that they may be rigidly secured in position on a stand. Covers for containers shall be seamless, not less than 0.032 inch thick, substantially hinged, arranged to seat properly, and provided with hasps and staples or other suitable means to lock them securely. A wire handle suitable for carrying the container and for opening the cover by means of a foot lever on a container stand shall be attached to each cover.

3.8.15.2 Container stands.- Container stands shall be suitable for the number and size of the containers, with or without heater as specified (see 6.1), rigidly constructed, and provided with four substantial rubber-tired wheels, unless means for securing to floor or deck is specified (see 6.1). Stands shall be fitted with an independent foot lever for each container arranged for opening the cover and automatically closing it when pressure on the lever is released. Pedal of the foot lever shall have nonslipping surface.

3.9 Type II, instrument sterilizers, pressure type.-

3.9.1 The requirements for type I sterilizers specified in 3.8 shall apply to the type II sterilizers, except that the requirements specified in 3.8.15 are not applicable.

3.9.2 Frame and trays for instruments.- A frame and two trays shall be provided and shall be of corrosion-resisting steel. The trays shall be provided with lifting handles and shall have bottoms of perforated metal or of wire mesh. The trays shall be not less than 2-1/2 inches deep, and shall have projecting flange or supports on the bottom that prevent articles from contacting unsterile surfaces.

3.10 Type III, instrument sterilizers, boiling type.-

3.10.1 Type III, instrument sterilizers shall conform to table II for the size specified (see 6.1). Sizes 2 and 4 shall be electrically heated. They shall heat water at normal level to boiling temperature within the time specified. The time required to reach boiling temperature shall be based on water at 70°F., and depth at least 1/2 inch over the top of the tray in lowered position. The ambient room temperature shall be not more than 104°F. nor less than 78°F.

3.10.2 Mounting.- Sterilizers shall be mounted as follows:

Size 2 - Shelf mounted (may be used as portable sterilizer, but shall have provision for shelf mounting).
Size 4 - Shelf or stand mounted, as specified (see 6.1).
Size 5 - Stand mounted.

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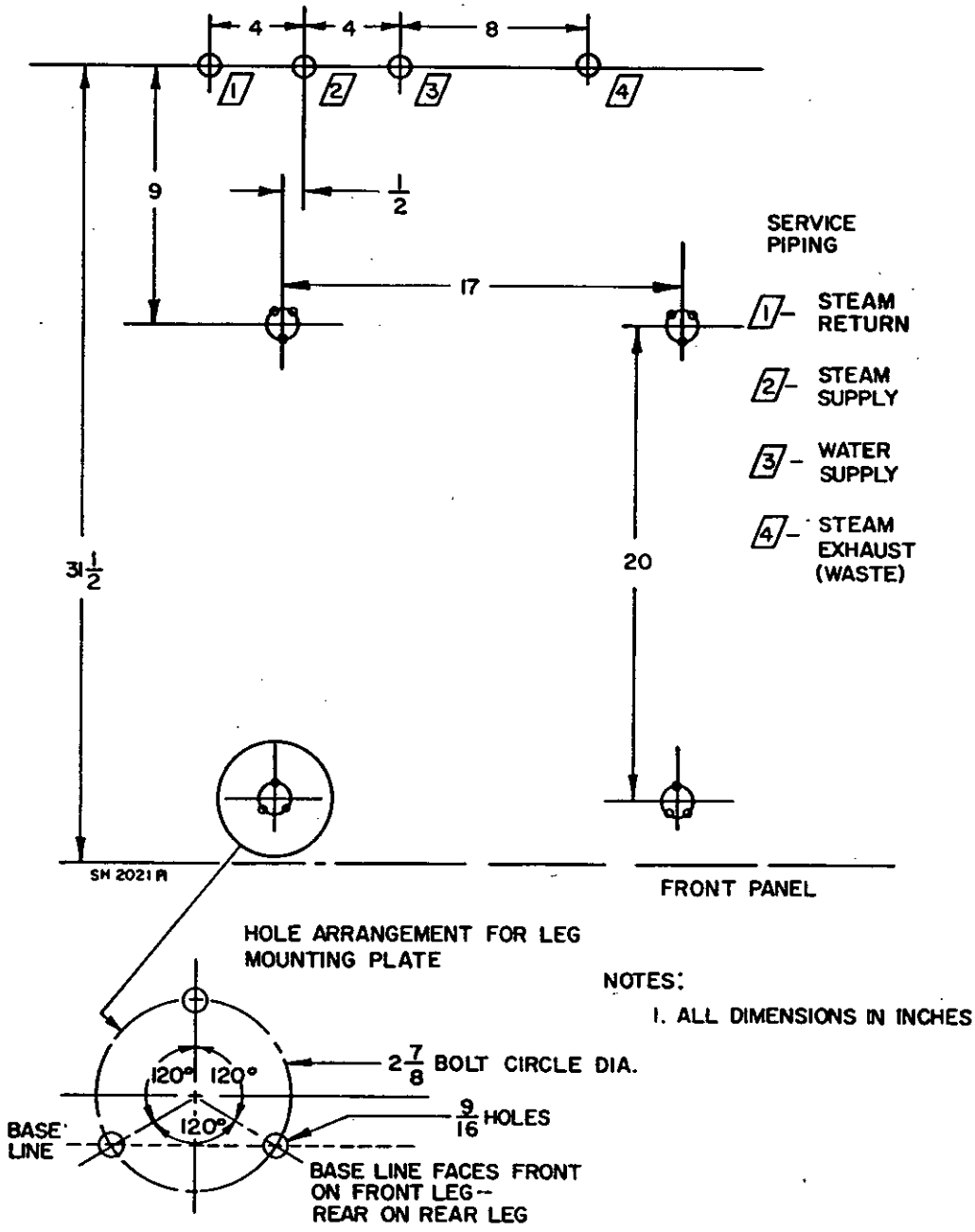


Figure 1 - Installation dimensions types I and II, size 2.

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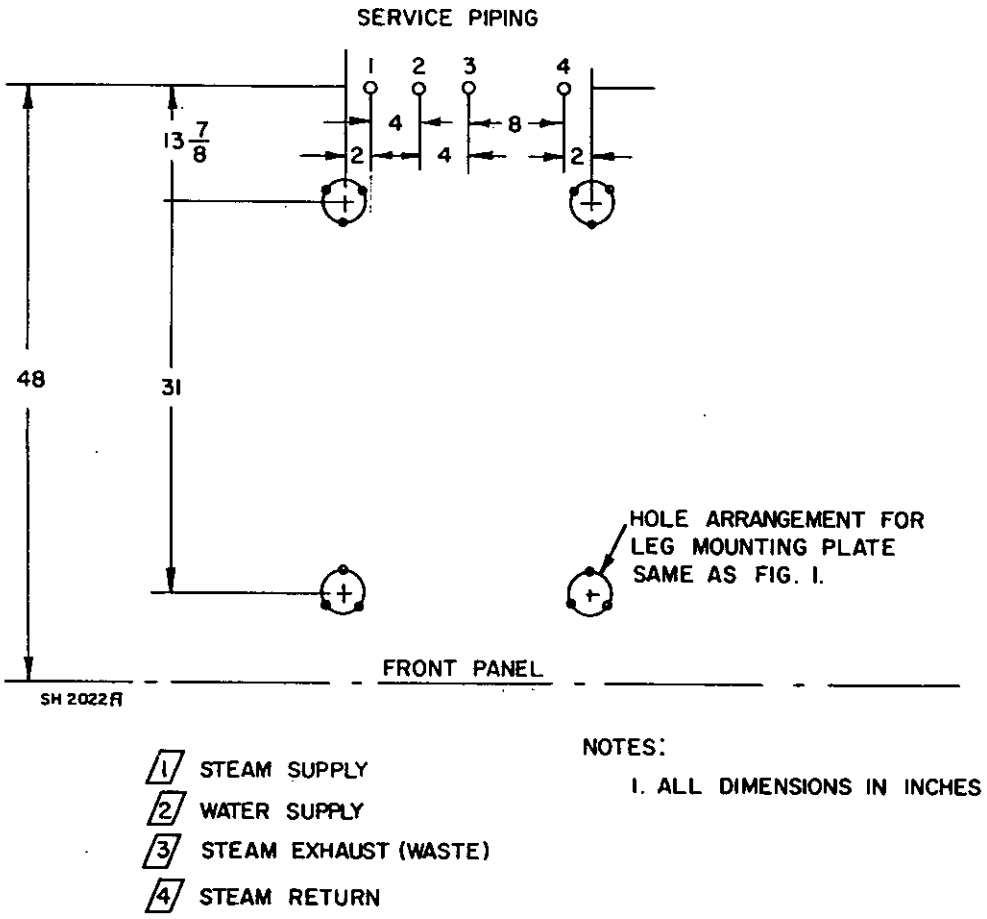


Figure 2 - Installation dimensions types I and II, size 3.

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Table II - Dimensions and requirements for type III, instrument sterilizers, boiling type.

Sizes	Nominal inside dimensions of tank			Thickness of material (minimum)			Time required to attain boiling temperature (maximum)		Power input (Maximum)
	Length	Width	Depth	Bottom	Sides	Cover	With steam heat	With electrical heat (at rated voltage)	
	Inches	Inches	Inches	Inch	Inch	Inch	Minutes	Minutes	KW
2	16	6	4	0.125	0.125	0.028	----	26	1.3
4	20	10	9	.035	.031	.035	----	30	5
5	22	12	10	.035	.035	.035	8	----	----

3.10.2.1 Shelf mounting. - Sterilizers shall be suitable for mounting on a shelf or counter top. Non-heat conducting feet shall be furnished on the bottom of the sterilizer. Tapped holes, drilled lugs or other means shall be provided for fastening the sterilizer securely in place. Plastic parts shall not be used for securing the mounting bolts, unless metal inserts are provided.

3.10.2.2 Stand mounting. - Sterilizers shall be mounted on a fixed floor stand of a convenient height.

3.10.3 Tanks. - The tanks shall be corrosion-resisting steel or nickel-copper alloy (monel). Seams shall be welded and joints ground smooth or plain locked and sweated. Corners shall be rounded smooth inside and top edges of tank shall be suitably reinforced.

3.10.4 Tank covers. - Tanks shall be fitted with a cover of the same material as the tank, arranged to swing clear of trays being handled. The cover shall be suitably reinforced and arranged to return all condensation to the tank and shall have heavy hinges finished to match the cover.

3.10.5 Tray. - Sterilizers shall be fitted with a full length tray of corrosion-resisting steel or nickel-copper alloy (monel). In addition, the size 2 sterilizer shall be fitted with a small basket of suitable size and of the same material as the tray to hold dental burrs or surgeon's needles. The trays shall be suitably reinforced, and rigidly constructed. Trays shall be provided with lift handles, such that in the closed position the trays shall rest within 1/2 inch of the heating coil or elements. Trays shall be constructed with a projecting flange or supports on the bottom which shall prevent sterilized articles from contacting unsterile surfaces.

3.10.5.1 Size 2 and the shelf mounted size 4 shall have a cover and tray which shall be raised simultaneously, automatically suspended in the open position, and lowered by a lever having a handle of non-heat- conducting composition. Size 4, when stand mounted, and size 5 shall have a foot-pedal device which shall simultaneously raise and automatically suspend the cover and tray in the open position, and lower them gradually.

3.10.5.2 The lifting mechanism shall be sturdy. The tray shall not tip over when unevenly loaded. The lift mechanisms shall be constructed of corrosion-resisting steel or nickel-copper alloy (monel). Pedals shall have non-slipping surfaces.

3.10.6 Water supply and waste, sizes 2 and 4. - Sizes 2 and 4 will be filled by hand pouring, and shall be provided with a draw-off cock.

3.10.7 Water supply and waste, size 5. - On size 5 sterilizers, provision shall be made for water supply and waste governed by conveniently located valves. A screen of corrosion-resisting steel shall be provided over the waste outlet.

3.10.7.1 The water connection shall be attached to the top of the tank and shall be provided with an air break, or air intake above the top of the tank, the air intake being so designed that it cannot be readily obstructed and will not permit the escape of steam. The connection shall admit water freely without splashing and shall provide against syphoning of the contents of the sterilizers back into the water-supply line.

3.10.7.2 Provision shall be made against leakage of water into the sterilizer through a leaking valve after the sterilizer is filled.

3.10.8 Steam heated sterilizers, size 5. -

3.10.8.1 The size 5 sterilizer shall be provided with steam coils of suitable area to produce a sterilizing temperature in the time specified. The coils shall be adequately secured in position.

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3.10.8.2 A vapor regulator shall be provided for the size 5 sterilizer which shall consist of a rugged and reliable automatic device for regulating the rate of heating by means of which the rate of heating shall be automatically reduced when the water begins to boil, so that just enough heat is supplied to keep the water boiling without wasting heat and without creating an excessive amount of waste vapor.

3.10.9 Electrically heated sterilizers, sizes 2 and 4. - The size 2 and 4 sterilizers shall be provided with electric heating elements having a wattage not greater than that specified in 3.10.1 (see table II).

3.10.9.1 Voltage. - Size 2 sterilizers shall operate on 115 volts, single phase a.c. Size 4 sterilizers shall operate on one of the other voltages specified in 3.5.2.

3.10.9.2 Two heats shall be provided, one to raise water to boiling temperature in the time specified in 3.10.1 (see table II) and the other to maintain mild boiling. The change in heats may be accomplished either by a manual switch or automatically by thermostatic means. For 440 volt, 3 phase sterilizers, a balanced 3 phase load shall be provided on both heats.

3.10.9.3 A low-water cut out shall be provided.

3.10.9.4 Electrical controls shall be suitably enclosed in the base of the sterilizer.

3.10.9.5 Cable and plug assemblies. -

3.10.9.5.1 Type III, size 2. - Sterilizers shall be provided with a cable and plug connector assembly. The cable shall be type HSJ conforming with ASA Publication C33.1 and shall extend from the sterilizer 10 feet plus or minus 6 inches. The plug connector shall be a parallel blade, 2 pole, three wire grounding type attachment plug cap conforming to ASA Publication C73a and shall be molded to the cable. The grounding conductor of the cable shall be attached to all exposed dead metal parts of the sterilizer at one end and shall be attached at the other end to the grounding blade wiring terminal of the attachment plug cap.

3.10.9.5.2 Type III, size 4. - Sterilizers for operation on 440 volts, 3 phase shall be provided with a cable and plug assembly conforming with Specification MIL-C-15987 except that the cable length shall be 10 feet plus or minus 6 inches. Sterilizers for operation on d.c. shall be suitable for permanent connection to the power source by the installing activity.

3.11 Type IV, water sterilizers, direct-steam and electric. -

3.11.1 Type IV water sterilizers shall have provision for being heated both from the direct steam supply and from an electric supply. They shall be capable of filtering water, heating it to a temperature of at least 250°F. in the tanks, maintaining this temperature for 20 minutes, and cooling the water in one of the two tanks. Each tank shall be equipped with a readily removable water filter, a steam coil, and an immersion type electric heating unit. One tank shall be equipped with a cooling coil.

3.11.2 Mounting. - Sterilizers shall be stand mounted with a corrosion-resisting steel panel adapted for an open or recessed installation. Gages and operating handles shall be located in front of the panel.

3.11.3 Tanks. - Sterilizers shall consist of two tanks, one for hot water, and one for cold water. Each tank shall have an identification plate marked "Hot" or "Cold", as applicable. Each water sterilizer tank shall conform to table III for the size specified (see 6.1). The time required to reach sterilizing temperature shall be based on water at 70°F. The ambient temperature shall be not less than 78°F. and not more than 104°F.

3.11.3.1 The tanks shall be mounted vertically and shall be such that the height of the sterilizer and the space to remove the shells shall not exceed 10 feet.

3.11.3.2 The shells shall be dome-shaped at the top and shall be constructed of seamless or welded corrosion-resisting steel or nickel-copper alloy (monel) and shall be polished.

3.11.3.3 The cold water tank shall be fitted with a cooling coil and valve which shall convey cooling water through the tank.

3.11.4 Filter. - A readily removable filtering device for each tank, with valve and connections for independent operation of each tank, shall be provided. Filters shall be mounted so as to be accessible from the front of the tanks, or from the operating side of the wall. Provision shall be made to protect the tanks against the leakage of raw water into the tanks after sterilization.

3.11.5 Draw-off faucets. - Separate draw-off faucets shall be provided for each tank. Faucets shall be bell-mouthed, with lever handles of non-heat conducting composition.

3.11.6 Drain basins. - There shall be provided underneath the faucets at a suitable level for holding a water pitcher, suitable drain basins of corrosion-resisting steel attached to stand or panel.

3.11.7 Waste piping. - The drain from the filter, the drain from each tank, and the drain from the cooling coils shall be piped to an open receiver(s), so arranged as to prevent back flow and contamination of

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Table III - Dimensions and requirements for type IV, water sterilizers.

Sizes	Capacity (Minimum)	Thickness of shell (Minimum)	Time required to attain sterilizing temperature (250°F.) (Maximum)		Power input total both tanks (Maximum)
			With steam heat	With electrical heat at rated voltage	
	Gallons	Inches	Minutes	Minutes	Kilowatts
1	8	0.058	15	45	22
2	15	.065	20	60	27.4

either the water supply or the contents of the sterilizer.

3.11.8 Fittings.- Each tank shall be supplied with the following fittings:

(a) An automatic air exhaust and air filtering vacuum breaking valve which shall open instantaneously at atmospheric pressure.
(b) A safety valve(s).
(c) A dial thermometer. The calibrated graduation shall be from not more than 110°F. to at least 280°F.
(d) A water level gage with means for automatic sterilization of water in the gage during the sterilizing process shall be provided.
(e) A pressure gage which shall have a range from 5 to 20 psig.

3.11.9 Steam heating.- Each tank shall be supplied with heating coils or tubes of suitable capacity to produce a sterilizing temperature in the time specified, and an automatic control which shall maintain the pressure within 2 psi of the set pressure, with a fluctuation in boiler pressure not to exceed 15 percent.

3.11.10 Electrical requirements.-

3.11.10.1 Voltages.- Voltages shall be in accordance with 3.5.2.

3.11.10.2 Wattage.- Wattage and heating time shall be in accordance with 3.11.3 (see table III).

3.11.10.3 Controls.-

3.11.10.3.1 A low-water cutout (see 3.5.5.4) shall be provided.

3.11.10.3.2 Each sterilizer shall be fitted with an automatic control which shall maintain the pressure within 2 psig of the set pressure, with a fluctuation in the supply voltage not to exceed 10 percent.

3.11.11 Installation dimensions.- The installation dimensions shall be as shown on figure 3.

3.12 Identification plate.- Each sterilizer shall be provided with a metal identification plate with lettering that is etched, engraved, stamped, cast or photoprocessed. The lettering shall be plainly visible, preferably black in color against a light background. The identification plate shall be securely fastened in a conspicuous location. The plate shall contain the following information:

(a) Name of manufacturer.
(b) Manufacturer's type designation.
(c) Serial number.
(d) Voltage and wattage rating (if electrically heated).
(e) Maximum and minimum steam pressure (if steam heated).
(f) Federal stock number, (if available).

3.13 Technical manuals.- Technical manuals shall be in accordance with type III of Specification MIL-M-15071. Unless otherwise specified (see 6.1), the quantity of manuals shall be in accordance with Specification MIL-M-15071. Each manual shall include a reduced size print of the approved master drawing.

3.14 Drawing.- Drawings shall be in accordance with Class A of Specification MIL-D-963 and shall contain the following minimum data:

(a) Top, front and side views of the sterilizer assembly with parts identified in the list of materials. The use of partial cross-section or cut-away views to show method of construction is encouraged.
(b) Mounting dimensions, overall dimensions and locating of piping connections.
(c) Detail of electrical heating unit assembly.
(d) Operating characteristics: wattage, voltage and steam pressure.
(e) Wiring diagram.
(f) Schematic piping diagram.
(g) Weight of sterilizer.

3.15 Onboard repair parts.- Onboard repair parts shall consist of the parts specified in table IV

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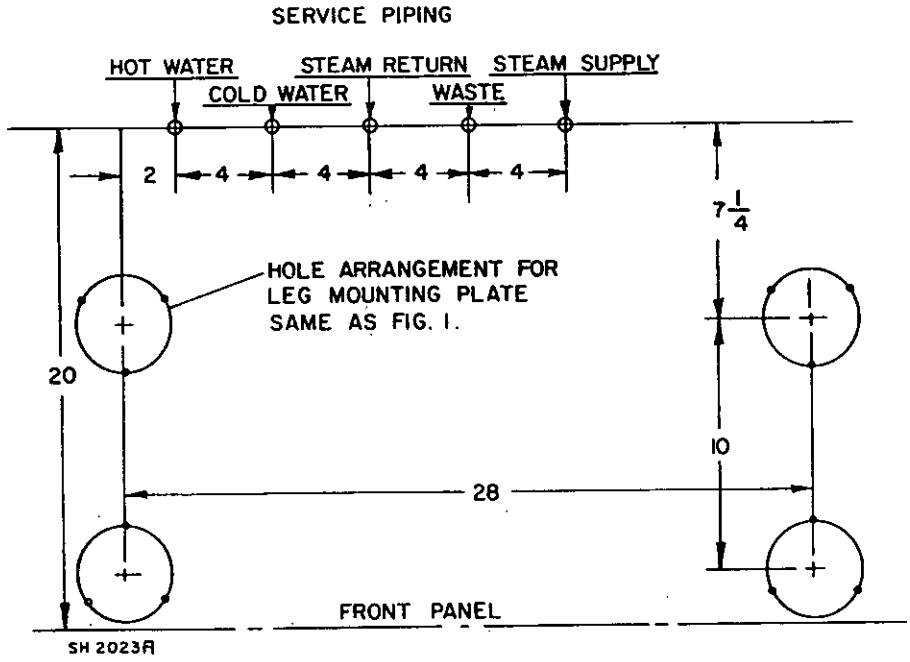


Figure 3 - Installation dimensions type IV, sizes 1 and 2.

and any additional parts considered by the manufacturer to be necessary for the maintenance of the sterilizer.

3.16 Workmanship. - Workmanship shall be first class in every respect and in accordance with the best commercial practice.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling for acceptance inspection. -

4.2.1 Lot. - All sterilizers of the same type, size and voltage offered for delivery at one time shall be considered a lot for purposes of acceptance inspection.

4.2.2 Sampling for group A examination and tests. - A random sample of sterilizers shall be selected in accordance with table V from each lot and shall be subjected to each of the group A examination and tests specified in 4.3.1.

4.2.3 Sampling for group B tests. - A random sample of sterilizers shall be selected from each lot in accordance with table VI and shall be subjected to each of the group B tests specified in 4.3.2.

4.2.4 Sampling for group C tests. - The first sterilizer of a particular type, size and design shall be subjected to the group C tests specified in 4.3.3. On subsequent contracts or orders for sterilizers of identical construction, the tests need not be repeated, provided that a copy of the original test report, certified by an inspector is made available.

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Table IV - Onboard repair parts

Name	Total number of identical parts installed per ship						
	1	2	3	4	5-8	9-20	Over 20
	Quantity of onboard repair parts to be furnished per ship						
Electric heating elements (each size and type)	1	1	1	1	2	3	4
Heating elements, gaskets	1	1	1	1	2	3	4
Valve discs (each size and type)	1	2	3	3	4	6	7
Steam trap, complete or steam trap parts consisting of: (a) thermosensitive element (b) Valve seat (c) Gaskets	1	1	1	1	2	3	4
Steam pressure regulator or regulator parts consisting of: (a) Diaphram assembly (b) Valve disc (c) Gasket	1	1	1	1	2	3	4
Door gaskets (types I and II only)	1	2	3	3	4	6	7
Steam generator gaskets	1	2	3	3	4	6	7
Water gage glasses	1	2	3	3	4	6	7
Control switches	1	1	1	1	2	3	4
Thermostatic switches (including low water cut outs)	1	1	1	1	2	3	4
Prssure switches	1	1	1	1	2	3	4
Filter, complete or filter replacement parts as recommended by manufacturer	1	2	3	3	4	6	7
Contactors or relays complete or contactor or relay repair parts in accordance with Specification MIL-R-2033, MIL-C-2174, MIL-C-2212, as applicable	1	1	1	1	2	3	4

Table V - Sampling for group A examination and tests
AQL (approx.) = 1.5 percent defective

Number of sterilizers in lot	Number of sterilizers in sample	Acceptance number	Rejection number
		Number of sterilizers nonconforming on any group A examination and test	
15 and under	7	0	1
16 to 40	10	0	1
41 to 110	15	0	1
111 to 300	25	1	2
301 to 500	35	1	2
501 and over	50	2	3

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Table VI - Sampling for group B tests

Number of sterilizers in lot	Number of sterilizers sample	Acceptance number	Rejection number
		Number of sterilizers nonconforming on any group B test	
15 and under	3	0	1
16 to 40	5	0	1
41 to 110	7	0	1
111 to 300	10	0	1
301 and over	15	1	2

4.3 Acceptance inspection. -

4.3.1 Group A examination and tests. - Each of the sterilizers selected in accordance with 4.2.2 shall be subjected to each of the examination and tests specified herein and the results of each examination and test compared with the requirements of this specification. Failure to conform to this specification for any group A examination and test shall be counted as a defect and the sterilizer shall be rejected. If the number of such nonconforming sterilizers in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

Group A examination and tests	Reference paragraph
Adjustment, fit and materials	4.4.1
Dimensions	4.4.2
Wattage	4.4.3
Dielectric (cold)	4.4.4
Insulation resistance (cold)	4.4.5
Pressure (types I and II)	4.4.6
Examination of repair parts	4.4.1.1

4.3.2 Group B tests. - Each of the sample sterilizers selected in accordance with 4.2.3 shall be subjected to each of the tests specified herein, and the results of each test compared with the requirements of this specification. Failure to conform to this specification for any group B test shall be counted as a defect and the sterilizer shall be rejected. If the number of such nonconforming sterilizers in any sample exceeds the acceptance number for that sample, the lot represented by that sample shall be rejected.

Group B tests	Reference paragraph
Operation	4.4.7
Dielectric (hot)	4.4.4
Insulation resistance (hot)	4.4.5

4.3.3 Group C tests. - Each of the sample sterilizers selected in accordance with 4.2.4 shall be subjected to each of the tests specified herein, and the results of each test compared with the requirements of this specification. If visual and dimensional examination reveals variations beyond manufacturing tolerance, the inspector may require that any or all of the group C tests be conducted on a particular sterilizer to demonstrate that it all conforms to this specification. No sterilizer shall be accepted for delivery until these tests have been satisfactorily completed. Failure of the sterilizer to pass any of the tests shall result in the rejection of all units so manufactured until the cause of rejection has been corrected on all sterilizers and the sample has been successfully retested. Sterilizers of the same type and design need not be retested on subsequent contracts or orders, therefore, a record of the test, certified by the inspector shall be retained by the manufacturer.

Group C tests	Reference paragraph
Reliability	4.4.8
Shock	4.4.9

4.4 Examination and tests

4.4.1 Adjustment, fit and materials. - The sterilizers shall be subjected to a thorough examination to ascertain that the materials, workmanship and design are in strict conformance with the requirements of this specification, contract and approved drawing. The fit of parts shall be observed with particular attention given to the interchangeability of such parts as are likely to require replacement during normal service life of the sterilizers.

4.4.1.1 Examination of repair parts. - All repair parts shall be examined to determine that the parts are duplicates of the parts furnished in the sterilizers. If the inspector has any reason to doubt the ready interchangeability of the repair parts with the original parts, he may require a demonstration of interchangeability. The wattage of the heating elements furnished as repair parts shall be measured in accordance with the sampling for group A examination and tests.

4.4.2 Dimensions. - The sterilizers shall be measured to determine that the mounting and overall dimensions conform with the requirements of this specification and the approved drawing.

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4.4.3 Wattage. - The wattage of the sterilizers at rated voltage shall be measured to determine conformance with tables I, II, and III.

4.4.4 Dielectric. - Electrically heated sterilizers shall be subjected to and shall withstand, for a period of one minute, a dielectric test voltage of twice the rated voltage plus 1000 volts applied between each circuit and the body of the sterilizer. During the test all other circuits and metal parts shall be effectively grounded and all relays or contactors shall be closed. Dielectric tests shall, in general, be made upon the completely assembled equipment and not upon individual parts. The dielectric test (cold) shall be conducted at normal room temperature. The dielectric test (hot) shall be conducted as soon as practicable after the operation test (see 4.4.7).

4.4.5 Insulation resistance. - The insulation resistance of electrically heated sterilizers shall be measured to determine conformance with 3.5.4.

4.4.5.1 The insulation resistance test shall be made with all circuits of equal voltage above ground connected together. Circuits or group of circuits of different voltages above ground shall be tested separately. Tests shall be made using a potential of 500 volts d.c.

4.4.5.2 The insulation resistance test shall be made under the following conditions:

(a) Insulation resistance (hot) - As soon as practicable following the dielectric test (hot).
(b) Insulation resistance (cold) - At approximately room temperature.

4.4.6 Pressure (types I and II only). -

4.4.6.1 The space between the inner and the outer shells and all pressure tanks shall be tested with a hydrostatic pressure of not less than 1.5 times the working pressure during which there shall be no leakage or sweating. There shall be no permanent set or deformation upon releasing pressure. Doors shall be steam tight under a test steam pressure of 1.5 times the working pressure.

4.4.6.2 The complete steam generator shall be subjected to a hydrostatic pressure of 50 psi for 10 minutes and shall show no leakage or sweating.

4.4.7 Operation. - The sterilizer and associated controls shall be operated under normal operating conditions for at least 2 hours to determine whether the sterilizer performs satisfactorily. Sterilizers that may be operated from either a direct steam or electric supply shall be operated for at least one hour from each. At the start of the test, the time required to reach sterilizing temperature shall be measured to determine conformance with 3.8.2, 3.9.1, 3.10.1 and 3.11.3. During the test there shall be no leakage from the valve or steam generator. The operation of components such as switches, indicator lights, thermometers, gages, safety valves, thermostats and low-water cut-outs shall be observed and shall be satisfactory and accurate. The operation of the protective devices such as safety valves and low-water cut-outs shall be accomplished by short circuiting of operating components, operating without water or otherwise simulating abnormal conditions which would result in operation of the protective devices.

4.4.8 Reliability. - Reliability shall be tested by operating the sterilizer for 25 complete sterilizing cycles under normal conditions in sterilizing full loads of dressings, instruments or water, as applicable. The sterilizing cycle for type I sterilizers shall include drying. The operation of the sterilizer and all components shall be satisfactory and reliable. When the same control settings are use, the operation of the sterilizer shall be consistent.

4.4.9 High-impact shock test. - Shock tests shall be conducted in accordance with Specification MIL-S-901. The features of the test shall be as follows:

(a) The required type of shock test. - Type A.
(b) The weight designation of the shock test. - As required by the sterilizer.
(c) Principal functions of the equipment. - Sterilizer.
(d) A definition of "failure to perform principal functions". -
  (1) Breakage of any parts including glass gage windows.
  (2) Appreciable distortion or dislocation of any parts.
  (3) Low dielectric strength. This dielectric strength test shall be made in accordance with 4.4.4, except that it shall be made with an applied voltage equal to 65 percent of that specified in 4.4.4. Under these conditions, insulation failures shall be cause for rejection.
  (4) Failure to operate satisfactory as determined from post-shock operation test conducted in accordance with 4.4.7.
(e) Description or designation of acceptable method of mounting the sterilizer on the shock-testing machine. - The mounting adapters shown on the following figures

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of Specification MIL-S-901 shall be used:

(1) Figure 4C for sterilizers to be tested on the shock-testing machine for lightweight equipment.

(2) Figure 7A for sterilizers to be tested on the shock-testing machine for medium weight equipment.

(f) The place at which the shock test will be conducted. - At the manufacturer's plant unless otherwise specified in the contract or order. Where a manufacturer is unable to conduct these tests at his own plant, he may arrange to have them conducted at a laboratory satisfactory to the Bureau of Ships, suitably equipped to conduct the tests.

(g) The number of sterilizers to be tested. - See 4.2.4.

(h) Disposal of shock tested sterilizers. -

(1) Sterilizers which have been subjected to the high-impact shock test and have failed to meet the requirements shall not be acceptable for Naval shipboard use.

(2) Sterilizers which have been subjected to the high-impact shock test and have successfully passed this test shall be considered acceptable under the contract or order after replacement or repair of any deformed or distorted parts.

4.5 Inspection of the preservation, packaging, and packing to verify conformance with the requirements of section 5 shall be performed in accordance with the sampling plans and inspection procedures of Specification MIL-P-116.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts. -

5.1.1 Sterilizers. -

5.1.1.1 Preservation and packaging. - Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical damage or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

5.1.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, contract or order number, contractor's name and destination.

5.1.2 Onboard repair parts. - Onboard repair parts shall be preserved and packaged level A, packed level C, and marked levels A and C in accordance with the item specification or Specification MIL-E-17555, as applicable. Items not covered therein shall be unit protected and packaged in accordance with the guide lines of Specification MIL-P-116. Items shall be individually packaged, except when used in sets.

5.2 Domestic shipment and storage or overseas shipment. - The requirements and levels of preservation, packaging, packing and marking for shipment shall be as specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

5.2.1.1 Preservation and packaging. -

5.2.1.1.1 Sterilizers. -

5.2.1.1.1.1 Level A. - Extent of disassembly of the sterilizers shall be in accordance with the recommendations of the equipment manufacturer subject to the approval of the Government inspector. Sterilizers shall be unit protected and packaged in accordance with method III of Specification MIL-P-116. Control knobs, gages, door handles and all other projections shall be cushioned to prevent damage during handling, shipment and storage. Cushioning material shall conform to Specification PPP-P-291 or PPP-C-843 and secured in place with gummed paper tape conforming to Specification UU-T-116 or equivalent. Doors, where provided, shall be closed and sealed. Each sterilizer shall be provided with a waterproof cover fabricated from barrier-material conforming to Specification UU-P-271, or minimum 0.003 inch thick polyethylene. Covers shall be sealed with tape as specified herein.

5.2.1.1.1.2 Component parts. - Trays, containers, container stands, recording thermometers and disassembled parts shall be individually unit

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protected and packaged in accordance with method III of Specification MIL-P-116 as follows:

(a) Contact preservatives shall not be required.

(b) The parts shall be cushioned or supplied by means of cellulosic materials of die-cut inner packing forms and unit packaged in paper or fiberboard boxes conforming to Specification PPP-B-566, PPP-B-665, PPP-B-676 and PPP-B-636 respectively at the option of the contractor. Box closures shall be in accordance with the appendix to the applicable box specification.

5.2.1.1.1.2.1 Component parts packaged as specified in 5.2.1.1.1.2 shall be placed in a snug-fitting fiberboard box conforming to class 2 of Specification PPP-B636 at the option of the contractor. All center and edge seams and manufacturer's joint shall be sealed with minimum 2-inch wide tape conforming to Specification PPP-T-60 or PPP-T-76 at the option of the contractor. The fiberboard box shall then be secured to the base or platform of the sterilizer shipping container, in an unused portion, with a minimum of two 5/8 inch by 0.015 inch steel straps conforming to class B of Specification QQ-S-781, or equivalent.

5.2.1.1.2 Repair parts.- Repair parts shall be preserved and packaged level A in accordance with Specification MIL-E-17555 or the item specification, as applicable. Items not covered therein shall be unit protected and packaged in accordance with the guide lines of Specification MIL-P-116. Unless otherwise specified, parts shall be individually packaged, except when used in sets.

5.2.1.2 Packing.-

5.2.1.2.1 Equipment and component parts.-

5.2.1.2.1.1 Level A.- Each sterilizer and component parts shall be packed for overseas shipment in containers conforming to any one of the following specifications at the option of the contractor:

Specification	Type or class
PPP-B-601	Overseas type
PPP-B-621	Class 2
PPP-B-636	Class 2
PPP-B-640	Class 2

Box closure, strapping or banding shall be in accordance with the applicable box specification or appendix thereto. Wood boxes exceeding 200 pounds gross weight shall be modified by the addition of skids in accordance with the applicable box specification. Fiberboard boxes shall not exceed the weight limitations of the fiberboard box specification.

5.2.1.2.1.2 Level B.- Each sterilizer and component parts shall be packed for domestic shipment and storage in containers conforming to any one of the following specifications at the option of the contractor:

Specification	Type or class
PPP-B-601	Domestic type
PPP-B-621	Class 1
PPP-B-636	Class 1
PPP-B-640	Class 1

Box closure shall be in accordance with the applicable box specification or appendix thereto. Wood boxes exceeding 200 pounds gross weight shall be modified by the addition of skids in accordance with the applicable box specification. Fiberboard boxes shall not exceed the weight limitations of the fiberboard box specification.

5.2.1.2.1.3 Anchoring, blocking, bracing and cushioning of equipment shall be provided and shall be in accordance with Specification JAN-P-100.

5.2.1.2.2 Repair parts.- Repair parts shall be packed level A or B as specified in accordance with Specification MIL-E-17555.

5.2.1.3 Marking.- In addition to any special marking required, interior packages and exterior shipping containers shall be marked for shipment in accordance with Standard MIL-STD-129 and the applicable packaging specification.

5.2.1.4 Technical manuals.- Technical manuals, pamphlets and handbooks, as applicable, shall be packaged, packed and marked for the level specified in accordance with Specification MIL-M-15071.

5.2.1.5 Drawings.- Drawings shall be packaged, packed and marked in accordance with Specification MIL-D-963.)

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type required (see 1.2).
(c) Voltage required (see 3.5.2).
(d) Size required (see 3.8.2, 3.10.1, and 3.11.3).
(e) Whether a recording thermometer is required for type I and II sterilizers (see 3.8.14).
(f) Whether containers for dressings are required for type I sterilizers (see 3.8.15).

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(g) Whether container stands are required with type I sterilizers (see 3.8.15) and whether the container stands, if required, shall have heaters and provision for being secured to the floor or deck, rather than having wheels (see 3.8.15.2).

(h) Whether type III, size 4 sterilizers shall be stand mounted or suitable for shelf mounting (see 3.10.2).

(i) Quantity of technical manuals required if quantities specified in 3.13 are not applicable.

(j) Preservation, packaging, packing and marking requirements other than those required in 5.1 (see 5.2).

Patent notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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